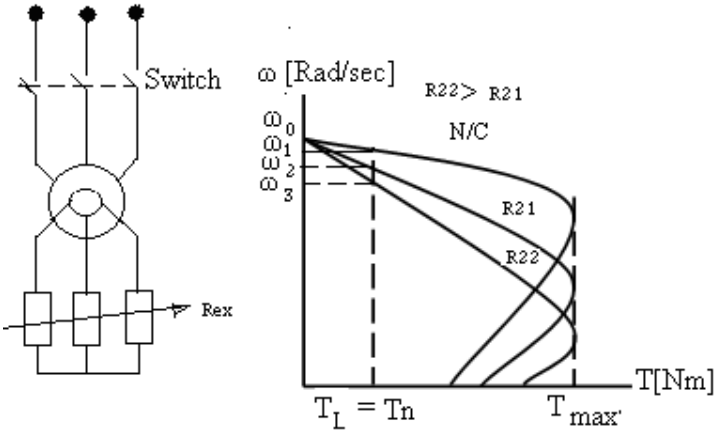
Unit Six

Slip Ring AC Motor Speed Control

These techniques are used for slip-ring (wound rotor) ac motors

1- **Inserting resistance into the rotor circuit.**



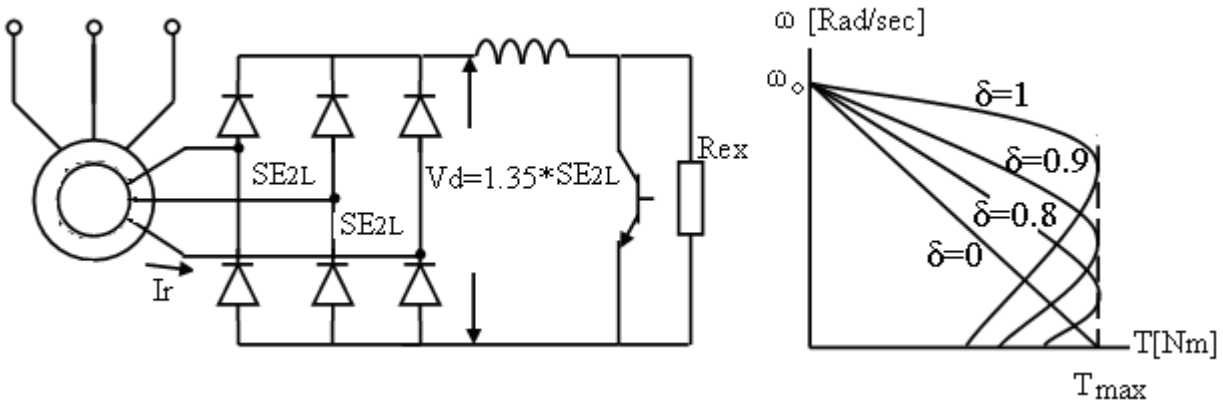
As we have seen from the equation of the maximum current, the maximum current decreases as we increase the resistance in the rotor circuit, at the same time, the starting torque as shown from the characteristics increases with the increasing of the rotor circuit resistance. This means that we can use this method for heavy start (start with the presence of large loads) and even we can start this motor with torque equals the maximum torque of the motor. Also, the load shouldn't be decreased because the maximum torque is constant and doesn't depend on the rotor circuit resistance.

$$I'_{2sc}=\frac{V_1}{\sqrt{(R_1+R'_{2\Sigma})^2+(X_1+X'_2)^2}}$$

The disadvantages of this method are:

1- Smoothness of speed control depends on the number of steps of resistance that are available.
2- Control is provided in the direction downward from the base speed.
3- Hardness decreases as the speed lowers.
4- Rang of speed control is not constant and depends on the value of the load (2: 1 to 3: 1).
5- A shortcoming of this method is the sizable energy losses

In order to dissipate energy in an external resistance, solid-state control of the external resistance can be achieved by the following circuit:



This will offer smoothness operation.

Slip energy in the rotor circuit is:

$\Delta P_2 = P_{em} - P_{out} = T\omega_o - T\omega = T(\omega_o - \omega) = T\omega_o s$

for the chopping:

δ = ton/tcycle -duty ratio of the chopper.

Effective resistance will be

$R_{ex}^* = (1-\delta)R_{ex}$:, so

$0 < R^*_{ex} < R_{ex}$ when $0 < \delta < 1$

Output voltage of the diode bridge is: $V_d = \frac{3SE_{m2L}}{\pi} cos0 = \frac{3S\sqrt{2}E_{2L}}{\pi} cos0$

$Vd = 1.35\ SE_{2L}$ where, E_{2L}- rms value of the line voltage

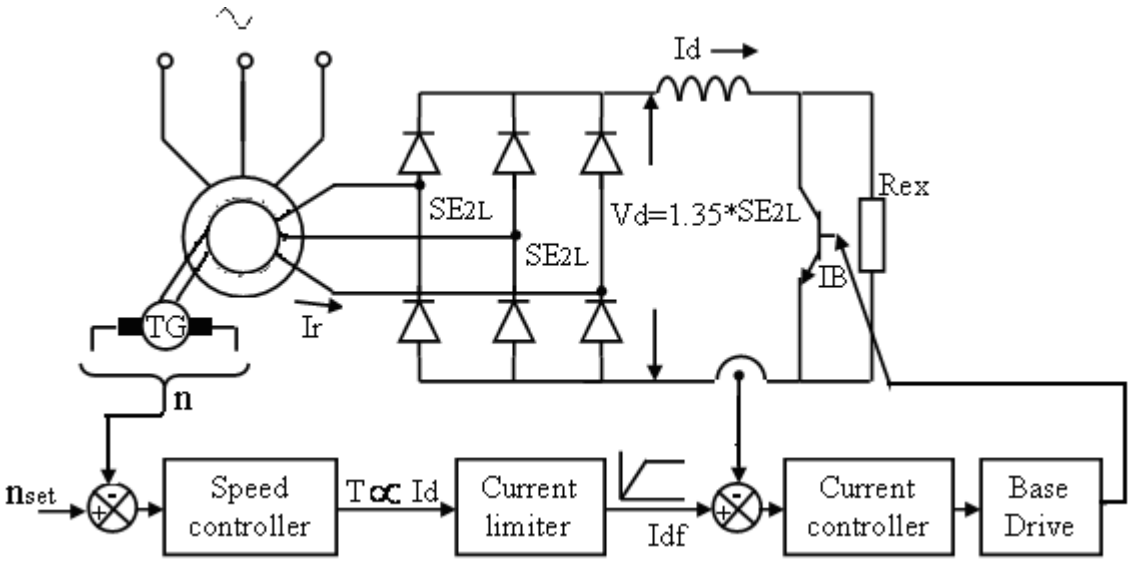
If we ignore losses in the rotor and consider all slip-energy dissipated in external resistance, then,

$T\omega_0 S = V_d I_d = 1.35 S E_{2L} I_d$

$$T = \frac{1.35E_{2L}}{\omega_0} I_d = KI_d$$

so, $T \alpha I_d$

Since, torque is proportional to the DC current, closed- loop system can be built where the actual speed is compared to a desired value and the error represents the reference current I_{df} can be compared to the DC current and the error will regulate the duty ratio- see the figure bellow:



$$I_r = \sqrt{\frac{2}{3}} \times I_d$$

Power consumed by R_{ex} per-phase:

$$P = \frac{I_d^2}{3} R_{ex}(1-\delta) = \frac{1.5 I_r^2}{3} R_{ex}(1-\delta) = 0.5 I_r^2 R_{ex}(1-\delta)$$

So, the rotor circuit resistance per-phase:

$R_{r\Sigma} = R_r + 0.5 R_{ex}(1-\delta)$,

Where:

Rex- external resistance

Example:-

A 440v, 50 Hz, 6 pole, Y connection, slip ring motor has the following parameters:

R1=0.5Ω, R2`=0.4Ω, X1=X2`=1.2Ω,Xm=5Ω

Stator to rotor turns ratio a=3.5, motor is controlled by static resistance control, external resistance is chosen such that the max torque is produced at stand still for duty ratio δ=0, calculate the value of external resistance (Rex) how δ should be varied with speed that the motor accelerates at max torque.

Solution:-

$$Scr = \frac{R2' + Rex}{\sqrt{R1^2 + (X1 + X2')^2}} = \frac{Rex + 0.4}{\sqrt{(0.5)^2 + (2.4)^2}} \Rightarrow Rex = 2.45.Scr - 0.4 \quad ...(1)$$

$$Rex = 0.5 * a^2 * Rex(1-\delta) = 0.5 * (3.5)^2 * Rex(1-\delta) = 6.125(1-\delta) * Rex \quad ....(2)$$

From equation (1) & equation (2):-

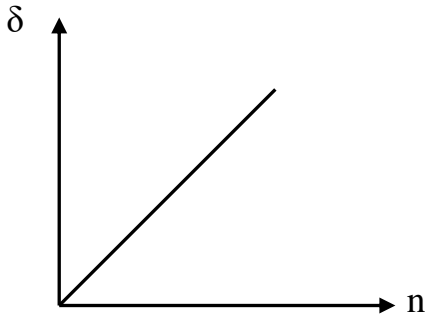
$$2.45 * Scr - 0.4 = 6.125(1-\delta) * Rex \quad ...(3)$$

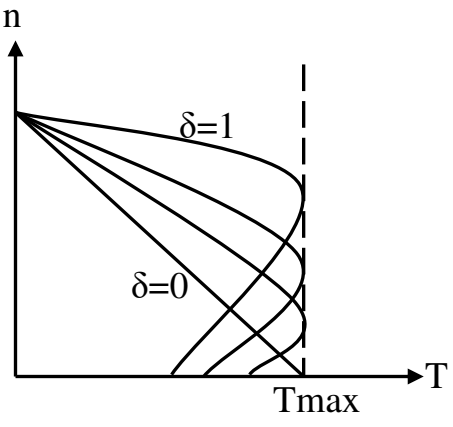
From max torque to occur at Scr =1 (stand still) for δ=0 :-

$$6.125 * Rex = (2.45 * 1) - 0.4 \Rightarrow Rex = 0.3347\Omega$$

Substituting the value of Rex and [S=1-n/n◦=(1-n/1000)] into equation (3):-

$$6.125 * (1-\delta) * 0.3347 = 2.45\left(1 - \frac{n}{1000}\right) - 0.4 \Rightarrow \delta = 1.195 \times 10^{-3} n$$





2. Speed Control of Slip Ring AC Motor using Slip Energy Recovery Schemes.

Slip losses are used to control the speed of slip ring ac motors by Cascade Methods, or the so-called (Slip Energy Recovery schemes).

A- Mechanical Cascade (Mechanical Scheme)

It's so named because it converts slip power into mechanical power on the motor shaft.

Figure (6-3) shows the mechanical cascade circuit.

Mechanical cascade components as follows:

1- Three-phase slip ring ac motor, where:

P_1 **: The input power of the motor, where:**

$$P_1 = 3V_1 I_1 \cos\varphi_1$$

P_2 **: Output power (load power) , where:**

$$P_2 = T_L \omega_L = T\omega$$

Under the influence of the slip power (ΔP_{2S}), a rotor electrical motive force appears on the rotor ends (rms phase voltage-as a source of electrical energy), which is directlyproportional to the slip value as follows.

$$E'_{2S} = E_{2n}.S$$

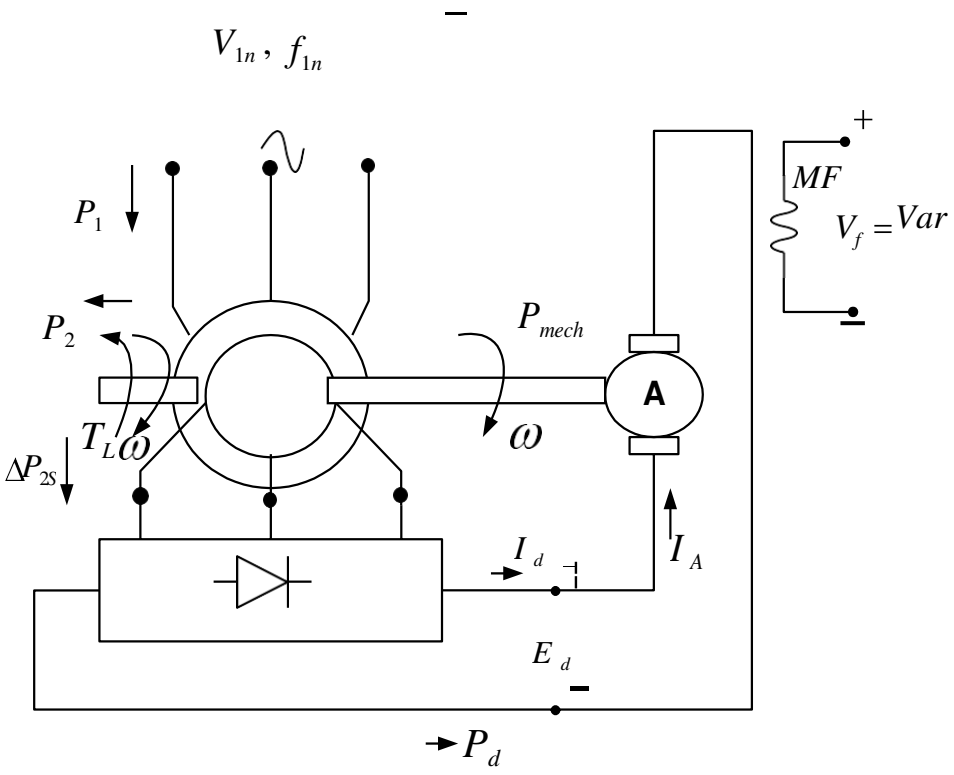


Figure (6-3) Mechanical cascade circuit

The rotor current (I'_2) will flow and its value equals:

$$I'_2 = \frac{E_{2n}.S}{R'_{2\Sigma}}$$

$R'_{2\Sigma}$: The total resistance of the rotor circuit.

(f'_2) Rotor frequency as AC power source frequency:

$$f'_2 = f_1.S$$

Thus, the rotor is an ac source for the uncontrolled three-phase rectifier.2- Three-phase full wave uncontrolled rectifier.

It converts AC power (ΔP_{2S}) into direct current power (P_d).

$$P_d = E_d I_d$$

Where:

I_d **: DC Link current.**

E_d **: DC link electric motive force and it equals:**

$$E_d = 2.34E'_{2S(PH)} = 2.34E_{2n}.S$$

The rectifier output is a feed source for the separately excited dc motor.

3- Separately excited dc motor.

It converts electrical power (P_d) **into mechanical power on the mechanical link that connects the armature of the dc motor and the rotor of the induction motor, and the two motors rotate at the same speed** (ω)**.**

Mechanical Power Value is:

$$P_{mech} = T_{DCM}\omega$$

Where:

T_{DCM} **: Torque of a direct current motor.**

The speed control of slip ring ac motor:

The value of the armature current (I_a) **equals to the dc link current** (I_d) **depends on the sum of the dc link electromotive force and the armature back electromotive force as follows:**

$$I_d = \frac{E_d - E_a}{R_\Sigma} = \frac{2.34.E_{2n}.S - K\phi\omega}{R_\Sigma}$$

Where:

$E_d = \frac{3\sqrt{3}V_m}{\pi} cos0 =$

$\frac{3\sqrt{3}\sqrt{2}V_s}{\pi} = 2.34V_s cos0 = 2.34sE_{2n}$, Vs-rms value of phase voltage

R_Σ **: The current path total resistance.**

The dc link current value is controlled by the variation of the dc motor magnetic flux value $(\phi = Var)$**, this is achieved by the dc motor field voltage variation** $(V_f = Var)$**.**

The cascade natural characteristic is when $(V_f = 0)$, then $(E_A = 0)$. That is, we get the highest value of dc link current equals to nominal value as follows.

$$\mathbf{I_d = I_{dn} = \frac{E_d}{R_\Sigma}}$$

Thus, the slip ring ac motor operates on the natural characteristic of mechanical cascade, at the nominal values (T_n , I'_{2n}).

In this case the speed control is achieved by varying field voltage as follows; $(V_f = V_{f1} > 0)$. And as a result $(E_a \neq 0)$, which causes a decrease of (I_d) and its value becomes less than nominal value $(I_d < I_{dn})$. This, in turn, affects the rotor current (I'_2), and it becomes less than the rated value $I'_2 < I'_{2n}$.

Accordingly, the electromagnetic motor torque becomes less than the load torque, and the resultant torque becomes negative $((T - T_L) < 0)$, according to the motion equation:

$$T - T_L = J\frac{d\omega}{dt}$$

Therefore, the motor will tend to reduce the speed. The slip increases with decreasing speed, so the slip power increases, which depends directly on the slip value, so the value of (I'_2) and value of (I_d) increase and the motor electromagnetic torque increases until the motor torque value is equal to the load torque $T = T_L$ as shown in figure (6-4):
Then the motor operates on regulated characteristic at speed (ω_B) as shown in Figure (6-4).

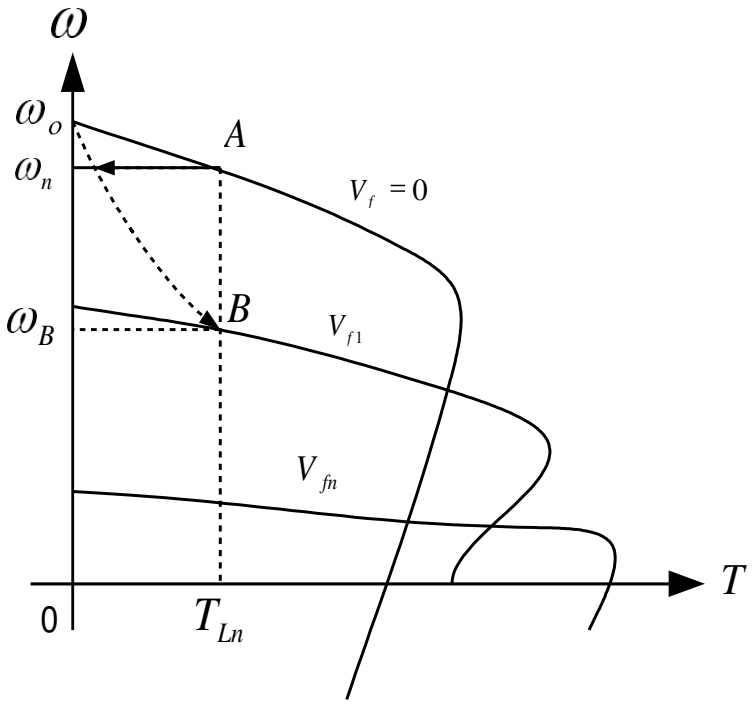


Figure (6-4) Mechanical characteristics of mechanical cascade

The transition from the natural characteristic to the regulated characteristic takes place in the first quadrant, because the rectifier does not allow the current to flow in the opposite direction (that is, the rotor current only passes in one direction). The transition characteristic is shown in dotted line on Figure (6-4). The final regulated characteristic is at a value $(V_f = V_{fn})$.

When the field voltage is not equal to zero $(V_f \neq 0)$, the dc motor will produce mechanical torque and mechanical power on the mechanical shaft.

$$T_{DCM} = K\phi I_A \neq 0$$

$$P_{mech} = T_{DCM}\omega \neq 0$$

This results in boosting the torque of the slip ring ac motor, so that the total torque on the motor shaft becomes:

$$T_{\Sigma} = T + T_{DCM}$$

The mechanical power on the motor shaft also increases, making the mechanical regulated characteristic shifted to the right as shown in Figure (4-6), so that the system works at constant power.

$$P_1 = P_2 + \Delta P_{\Sigma} \approx P_2 + \Delta P_{2S}$$
$$= T\omega + T\omega_o S = T\omega_o = \text{const.}$$

The slip power is converted into direct current power at the output of the rectifier and then into mechanical power on the mechanical shaft by the dc motor.

B- Electrical Cascade (Electrical Scheme)

It is called electrical cascade because it converts the electrical slip power into electrical power that feeds the source.

The electric cascade is divided into two types in terms of the principle of operation:

A - The Static Kramer Drive System.
And it regulates the speed below the natural characteristic (Sub synchronous Speed Control).

Figure (6-5) shows the electrical cascade circuit which is called The Static Kramer Drive System.

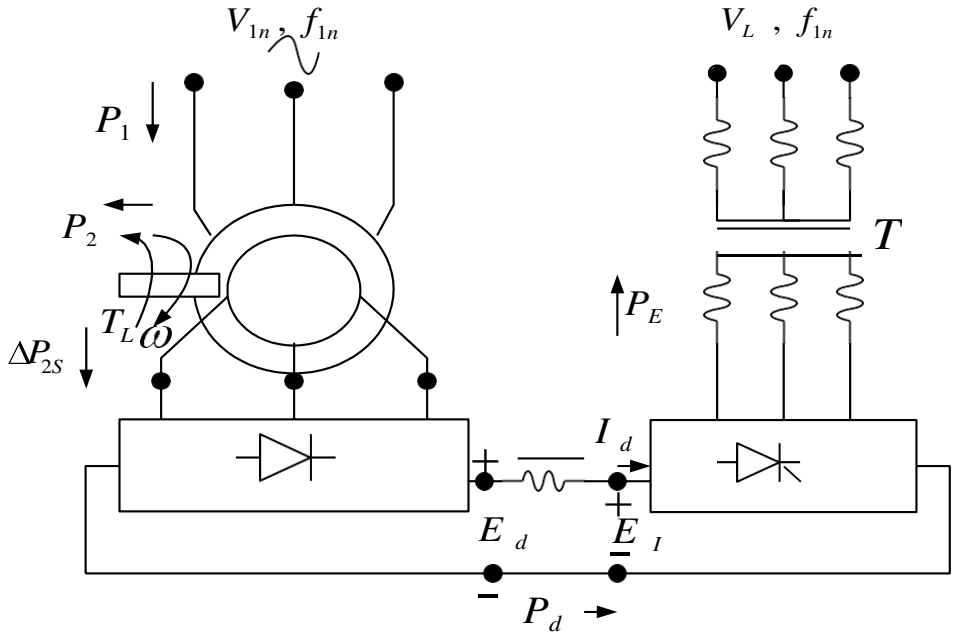


Figure (6-5) The Static Kramer Drive System

Electrical Kramer drive system components are:

1- Three-phase slip ring ac motor. On the rotor appears the slip power (ΔP_{2S}) as source for uncontrolled rectifier which parameters are; the electric motive force (E'_{2S}), current (I'_2) and frequency (f_2').

2- Three-phase full-wave uncontrolled rectifier

It converts the AC power (ΔP_{2S}) into direct current power (P_d).

3- Three-phase controlled inverter. It converts the direct current power into AC power (P_E). And it is transferred to the source by means of a star connected three-phase transformer, and it works to convert the voltage. And also perform the protection function (isolation between the source and the inverter).

Slip ring ac motor speed control operation:

DC link current (I_d) equals: $I_d = \frac{E_d - E_I}{R_\Sigma}$

Where:

E_d **: The dc link electromotive force.** $E_d = 2.34E_{2n}.S$

E_I **: The inverter electromotive force.** $E_I = 2.34E_{2T}.|COS\alpha|$

E_{2T} **: The transformer secondary electromotive force .**

α **: Inverter firing angle.** $(90° < \alpha < 180°)$

So, dc link current is:

$$I_d = \frac{2.34E_{2n}.S - 2.34E_{2T}.|COS\,\alpha|}{R_\Sigma}$$

When $(\alpha = 90°)$**, then:**

$$I_d = \frac{2.34E_{2n}.S}{R_\Sigma} = I_{d\,max} = I_{dn}$$

In this case, the motor operates on the natural characteristic of electrical cascade, as shown in Figure (6-6).

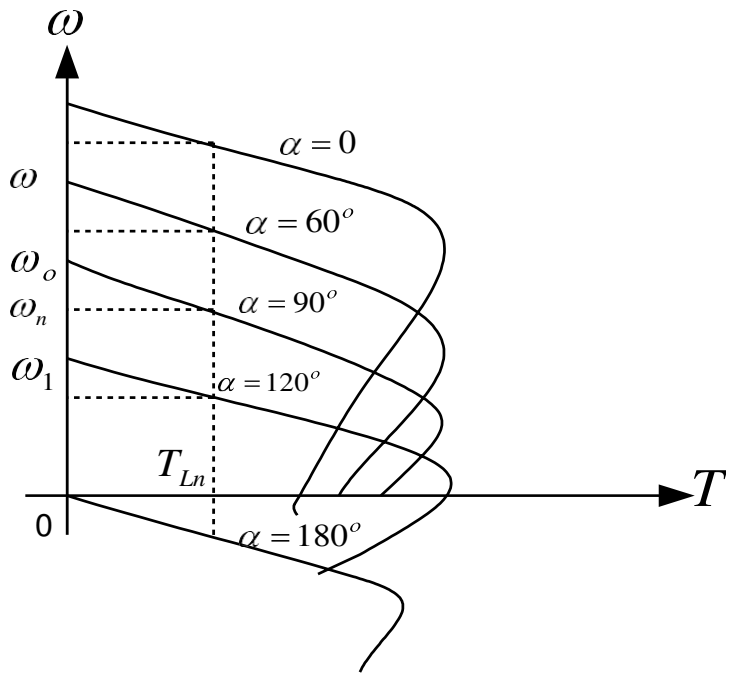


Figure (6-6) The static Kramer Drive System torque-speed characteristics

For example, when $(\alpha = 120^\circ)$ an electric motive force $(E_I > 0)$ appears, which causes a decrease of (I_d) and thus a (I_2') decrease, as well as the motor electromagnetic torque , which will lead to slowing down of the motor and the transition to a regulated characteristic below the natural characteristic. And it operates at load torque and speed (ω_1) The final regulated characteristic we get at $(\alpha = 180^\circ)$ as shown in figure (6-6).

Whereas, $(T_{max} = const.)$,the speed control in this case is performed with the constant torque load.

$$T_L = T_{Ln} = const.$$

B- Scherbuis Drive System.

The electrical cascade in this case is the same as Kramer drive system cascade, but a three-phase controlled rectifier is used when $(0 < \alpha < 90^\circ)$.

The slip ring ac motor has is a double- fed rotor

DC link current is:

$$I_d = \frac{2.34E_{2n}.S + 2.34E_{2T}.|COS\,\alpha|}{R_\Sigma}$$

The drive system operates on the natural characteristic when $(\alpha = 90^\circ)$.

For example, when $(\alpha = 60^o)$ an electric motive force of the rectifier $(E_R > 0)$ appears, which causes a increase of (I_d) and thus a (I_2') increases, as well as the motor electromagnetic torque , which will lead to acceleration of the motor and the transition to a regulated characteristic above the natural characteristic. And it operates at load torque and load speed.

The final regulated characteristic we get at $(\alpha = 0^\circ)$ as shown in figure (6-6).

When using an electrical cascade, the speed can be regulated in two directions above and below the natural characteristic. Thus, we obtain a wide range of speed regulation with the constant torque of the load at the specified value.